Temporal Topologies: Möbius Time, Circular/Helical Time, and Fractalized "Birthing" – A Unified Decimated-Transfer and Complex-Base Framework

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Abstract

We merge two complementary lines: (i) a *decimated transfer-matrix* description of spinor dynamics on a *temporal* Möbius cycle (APBC) versus a circular cycle (PBC), and (ii) a *complex-base* (s, b, l) framework that realizes self-similar (fractal) temporal envelopes and multi-track spawning. The result is a minimal, falsifiable picture of *net–zero yet infinite* time on the Möbius stage, a binary holonomy signature (the V_l sign flip), exact Green's functions and release laws (circular vs logarithmic spiral), and a mathematically controlled route to *fractalization/birthing* of local temporal cycles with convergence guarantees.

1 Temporal Möbius structure (properties and test)

1.1 Definition and holonomy

Definition 1.1 (Temporal Möbius cycle)**.** A time macrocycle of duration T contains $L \in \mathbb{N}$ Floquet steps. A *circular* (periodic) time cycle imposes $\psi(t+T) = +\psi(t)$ (PBC). A *Möbius* (temporal) cycle imposes $\psi(t+T) = -\psi(t)$ (APBC), i.e. an orientation flip in time; space is not twisted.

Let $U \in \mathrm{GL}(2, \mathbb{C})$ be the one-step (Floquet/transfer) update on a spinor. Over l steps define

$$V_l := \mathrm{tr}(U^l), \qquad Q_l := \det(U^l). \tag{1}$$

Cayley–Hamilton on U^l gives for any component x_m sampled every l steps the *universal local recurrence*

$$x_{m+1} - V_l\, x_m + Q_l\, x_{m-1} = s_m, \tag{2}$$

with source s_m (injected "ticks"). In closed SU(2) settings $Q_l = 1$ and $|V_l| \le 2$. The temporal holonomy shifts the single-step eigenphase $\theta \mapsto \theta + \frac{\pi}{L}$ (APBC). If the macrocycle length equals the decimation stride ($L = l$), then

$$\boxed{V_l^{\mathrm{APBC}} = -\,V_l^{\mathrm{PBC}}, \qquad Q_l^{\mathrm{APBC}} = Q_l^{\mathrm{PBC}}.} \tag{3}$$

Equation (3) is a *binary, falsifiable* signature of a temporal Möbius holonomy.

1.2 Exact Green's functions; net–zero yet infinite time

Let $r_\pm$ be the roots of $z^2 - V_l z + Q_l = 0$. The homogeneous impulse response G_n of (2) is

$$\text{distinct } r_\pm: \quad G_n = \frac{r_+^n - r_-^n}{r_+ - r_-},\ n \ge 0, \tag{4}$$

$$\text{double root } r: \quad G_n = n\, r^{\,n-1},\ n \ge 0. \tag{5}$$

In SU(2), set $V_l = 2\cos\varphi$ (so $r_\pm = e^{\pm i\varphi}$), then $G_n = \sin(n\varphi)/\sin\varphi$ and the forced solution is the exact convolution $x_m = \sum_{j<m} G_{m-j} s_j$ plus a two-term transient. Let $s(m) = (-1)^m$ be the temporal orientation along macro-steps. Define signed and total-variation times $\mathcal{T}_{\text{signed}}(M) = \sum_{m<M} s(m)\,T$, $\mathcal{T}_{\text{var}}(M) = MT$. Then for even $M = 2N$, $\mathcal{T}_{\text{signed}}(2N) = 0$ and $\mathcal{T}_{\text{var}}(2N) = 2N\,T \to \infty$. Moreover, for any APBC-odd linear readout $y_m = s(m)\,x_m$, the even-window Cesàro average vanishes while a Poisson register of ticks diverges linearly: this is *net–zero yet infinite time*.

1.3 Spectral check and Pin$^-$ remark

On a time circle of length T, the Euclidean Dirac operator $D = \frac{d}{d\tau}$ with boundary $\psi(\tau+T) = e^{i\pi s}\psi(\tau)$ has spectrum $\omega_n = \frac{2\pi}{T}(n + \frac{s}{2})$. PBC ($s$=0) has a zero mode; APBC (s=1) is gapped by π/T. Thus APBC removes the boundary zero mode (Pfaffian sign ambiguity) and is the unique reflection-positive choice for a temporal Möbius cycle, consistent with the $^-$ picture (details omitted here for brevity).

2 Circular and helical temporal structures

2.1 Circular (PBC) release: bounded echo

Post-quench (APBC→PBC) with $Q_l = 1$, $V_l = 2\cos\varphi$, the homogeneous solution is $x_{m_\star+n} = y_1\,U_{n-1}(\cos\varphi) - y_0\,U_{n-2}(\cos\varphi)$, a bounded echo ($U_n$ Chebyshev of the second kind). No radial drift.

2.2 Helical/spiral envelopes: two realizations

Physical (open/active). Allow openness $Q_l = q \neq 1$. Then $r_\pm = \sqrt{q}\,e^{\pm i\varphi}$ and $x_{m_\star+n} = \sqrt{q}^{\,n}(Ae^{in\varphi} + Be^{-in\varphi})$, a logarithmic spiral with radial factor $\sqrt{q}$ per stride and pitch $p = \frac{2\pi}{\varphi}\log\sqrt{q}$.

Geometric (complex-base embedding). Embed the decoded sequence in a complex base $B = b\,e^{i(2\pi/l)}$ ($b > 1$). Each stride-l step advances angle by 2π and scales radius by b^l; the drawing is a log-spiral with pitch $l\log b$. This is a *representation* layer; the dynamical tri-site coefficients remain the measured V_l, Q_l.

3 Fractalization / birthing via (s, b, l)

3.1 Complex-base scaffold (geometry/representation)

Fix integers $l \geq 2$, $b > 1$, $s \in (0, 1]$, and $U = e^{i2\pi/l}$, $B = b\,U$. A hierarchical value is

$$z \;=\; \sum_{\ell\geq 0} s^\ell \sum_{k\in\mathbb{Z}} d_{\ell,k}\,B^k, \tag{6}$$

with per-level finite support and absolutely summable tails. For quadratic b with $b^2 - Pb + Q = 0$, the *phase-aligned* (tri-site) identity on residue tracks is

$$B^{k+2l} - V_l(P,Q)\,B^{k+l} + Q^l\,B^k = 0, \tag{7}$$

(V_l a Lucas/Dickson polynomial). This identity generates local, value-preserving carries and yields canonical normal forms trackwise. *Caveat:* (7) is a *representation* identity (geometry), distinct from the *dynamical* tri-site law (2). We exploit it to design self-similar temporal envelopes and multi-scale spawns.

3.2 Track-dependent self-similarity

Choose a global similarity exponent M and tracks with $l_r \mid M$, bases $B_r = b^{M/l_r} e^{i2\pi/l_r}$; one full turn multiplies radius by $\beta = b^M$ on all tracks. The carry identity becomes track-uniform with coefficients $V_M(P,Q), Q^M$, enabling consistent multi-track normalization.

3.3 Birthing (spawn) rule and convergence

Let R_ℓ be a level-ℓ register (ticks accrued at level ℓ). A simple spawn law:

> If $R_\ell \geq R_c^{(\ell)}$, create a child level $\ell{+}1$ with scale $s^{\ell+1}$, choose a subset of tracks $\{l_r \mid M\}$, seed its initial digits from a linear map of the parent state, and reset $R_\ell \mapsto (1-\alpha_\ell)R_\ell$.

Termination (sufficient). If $\sup_\ell \mathbb{E}[\text{active digits at level } \ell] \leq C$ and $s < 1$, then $\sum_\ell s^\ell C$ converges; the hierarchical sum z is absolutely convergent. Thus fractal spawning produces a *finite* envelope while allowing arbitrarily many levels.

3.4 Coupling to decimated dynamics

At each active level, the decimated law (2) holds with the *measured* (V_l, Q_l) of the medium (per mode). Spawning sets initial conditions for child levels; the event-coded *pair* of checksums $I_\pm = \sum_k d_k r_\pm^k$ remains exact per mode (spinor-safe), providing robust counting across levels.

4 Binary tests, protocol, and decoding

1. **Holonomy flip.** Calibrate (V_l, Q_l) under PBC, switch to APBC (temporal Möbius), refit near steady-mode. For $L{=}l$, verify $V_l^{\text{APBC}} = -V_l^{\text{PBC}}$ and Q_l unchanged.
2. **Circular vs spiral.** Post-quench, fit Q_l. If $Q_l = 1 \Rightarrow$ bounded echo; if $Q_l \neq 1 \Rightarrow$ spiral with pitch $p = \frac{2\pi}{\varphi} \log \sqrt{Q_l}$.
3. **Fractalization.** Detect multi-scale spawns by level histograms in the (s,b,l) scaffold; convergence follows from the s-weighted support bound.
4. **Checksums (spinor-safe).** Use $I_\pm$ with $r_\pm$ roots of $z^2 - V_l z + Q_l = 0$; for SU(2), $|r_\pm| = 1$, so use both real/imag parts. A soft trellis adds penalty $\mu(|I_+(d) - I_+^{\text{ref}}|^2 + |I_-(d) - I_-^{\text{ref}}|^2)$.

Companion figures

Discussion

The temporal Möbius stage (APBC) enforces zero oriented DC while allowing unbounded accumulation on the cover. A quench to circular/spiral releases events with exact envelope laws and a binary signature (V_l flip). The (s,b,l) scaffold supplies a *geometric* mechanism for self-similar envelopes and controlled spawning with clear convergence conditions. The three pieces fit: local dynamics (decimated law), global holonomy (PBC vs APBC), and geometric self-similarity (complex-base tracks).

References (core pointers)

Witten (RMP 2016); Freed–Hopkins (arXiv:1604.06527); Kapustin et al. (JHEP 2015); Economou, *Green's Functions in Quantum Physics*; Rivlin, *Chebyshev Polynomials*; Kitagawa et al. (PRB 2010); Rudner et al. (PRX 2013).

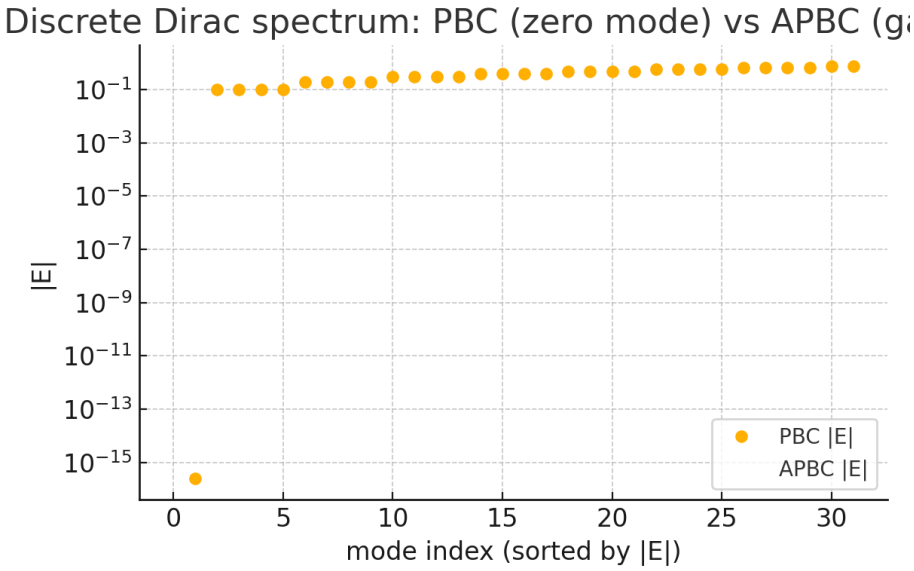

Figure 1: Discrete Dirac on a time circle: PBC shows a zero mode; APBC is gapped.

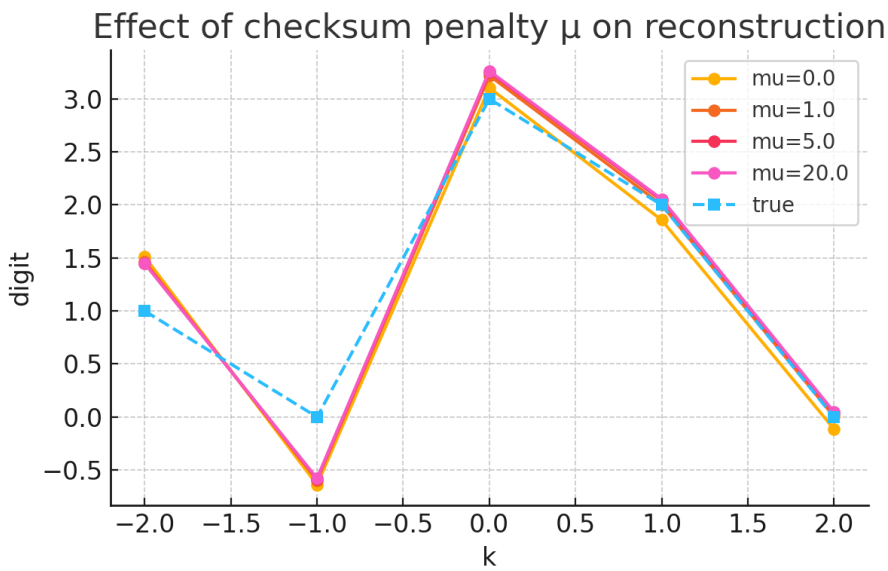

Figure 2: Checksum penalty μ stabilizes reconstruction on SU(2) roots.